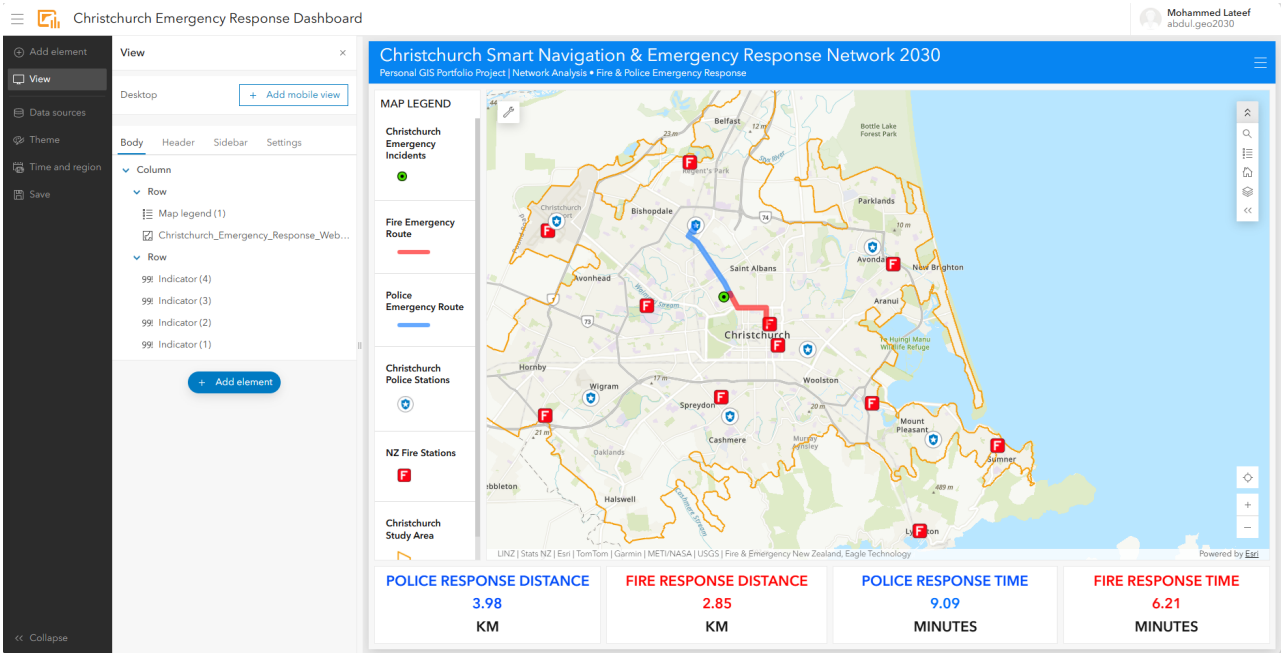


CHRISTCHURCH SMART NAVIGATION & EMERGENCY RESPONSE NETWORK 2030

Personal GIS Portfolio Project | Christchurch, New Zealand



ROLE	AUTHOR	DATE
GIS Analyst	Mohammed Abdul Lateef	September 2026

PROJECT OVERVIEW

This project demonstrates an end-to-end emergency response GIS workflow for Christchurch. A routable road network was prepared in ArcGIS Pro, travel-time impedance was configured, ModelBuilder was used to automate preparation, QA/QC and emergency response analysis, and the results were published to ArcGIS Online as an interactive web map and dashboard.

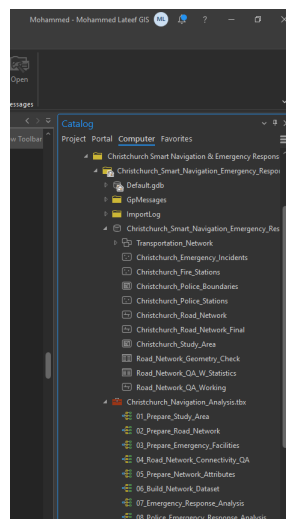
Objective: identify optimized fire and police response routes and communicate travel time, route distance, incidents and emergency facilities through a clear Web GIS application.

Core technologies: ArcGIS Pro | Network Analyst | ModelBuilder | ArcGIS Online | ArcGIS Dashboards | GIS QA/QC

MODELBUILDER AUTOMATION - 8-MODEL WORKFLOW

Eight ModelBuilder models were created to structure the project from data preparation and QA/QC through network construction and separate fire/police emergency response analysis.

Model	Purpose
01_Prepare_Study_Area	Prepares the Christchurch project boundary and defines the geographic extent used throughout the workflow. It provides a consistent study area for subsequent processing and analysis.
02_Prepare_Road_Network	Prepares and cleans the Christchurch road network for routing and network analysis. The output forms the transportation source required for the emergency-response network.
03_Prepare_Emergency_Facilities	Prepares fire and police station datasets as emergency-response facilities. It ensures facility locations are organized and ready for loading into network analysis.
04_Road_Network_Connectivity_QA	Performs QA/QC checks on road geometry and connectivity before network construction. It helps identify geometry or connectivity problems that could affect route solving.
05_Prepare_Network_Attributes	Prepares network attributes such as road length, speed and travel time for routing. These attributes support time-based impedance and emergency-response calculations.
06_Build_Network_Dataset	Builds the Christchurch Emergency Network Dataset from the prepared transportation network. The completed network becomes routable and ready for Network Analyst operations.
07_Emergency_Response_Analysis	Automates Closest Facility analysis by loading emergency facilities and incidents and solving the network. It identifies the optimized fire response route based on travel time.
08_Police_Emergency_Response_Analysis	Runs the police-specific emergency response workflow using the prepared network and police station facilities. It identifies the closest police response route and calculates response time and distance.



ArcGIS Pro Catalog showing the Christchurch Navigation Analysis toolbox and all eight ModelBuilder models.

NETWORK ANALYSIS & RESPONSE RESULTS

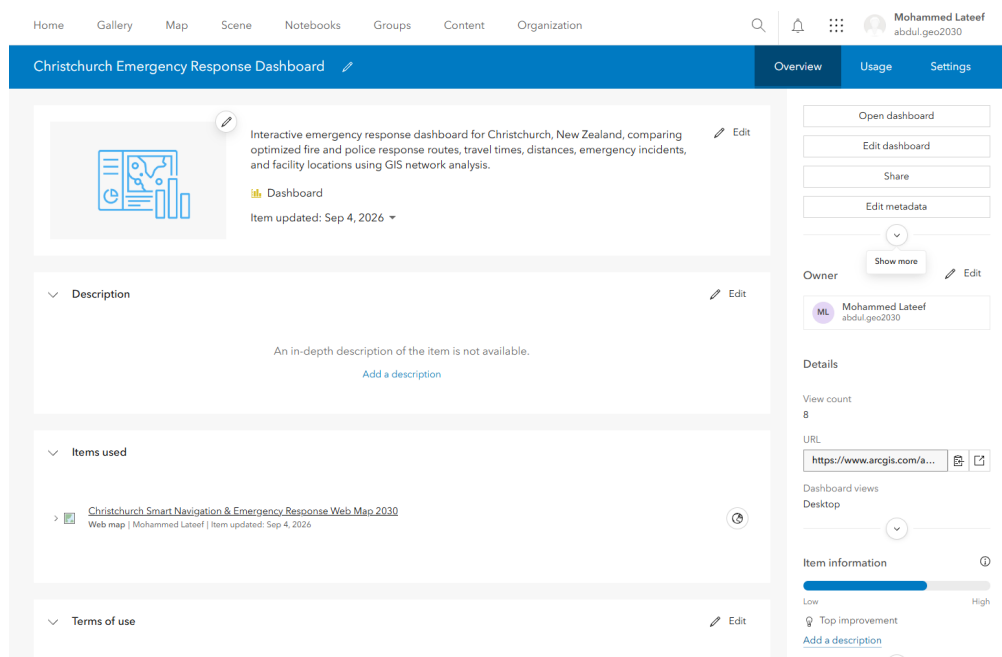
The road network was configured with a minute-based **TravelTime** cost using the **Travel_Time_Min** road attribute. After validation and network building, Closest Facility analysis was used to compare optimized fire and police response to incident INC_001.

FIRE RESPONSE	POLICE RESPONSE
6.21 min Response Time	9.09 min Response Time
2.85 km Route Distance	3.98 km Route Distance

ARCGIS ONLINE WEB MAP

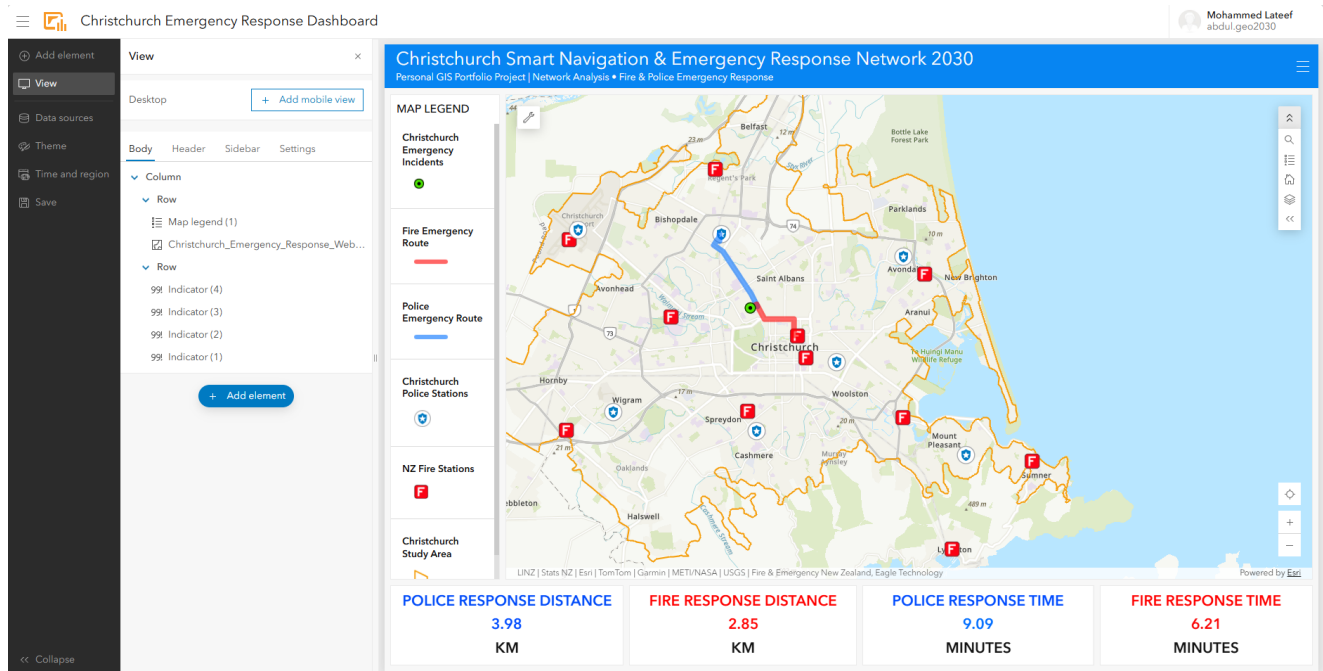
Analysis outputs were published to the **Christchurch Smart Navigation & Emergency Response Web Map 2030**. Pop-ups and symbology were configured to distinguish the emergency incident, fire route, police route, emergency facilities and Christchurch study area.

The web map was then connected to ArcGIS Dashboards to provide an operational view with response KPIs and interactive map controls.



ArcGIS Online item page showing the dashboard and its connected web map.

INTERACTIVE DASHBOARD & PROJECT OUTCOME



FINAL DASHBOARD

The completed dashboard combines an interactive emergency-response map, a map legend and four KPI indicators. Users can inspect route and incident pop-ups, search the map, zoom to locations and compare fire and police response performance.

KEY SKILLS DEMONSTRATED

Geodatabase preparation | Road-network QA/QC | Network Dataset configuration | Travel-time impedance | ModelBuilder automation | Closest Facility analysis | Fire and police emergency-response routing | ArcGIS Online publishing | Web-map configuration | ArcGIS Dashboard design

PROJECT OUTCOME

The project delivers a complete workflow from desktop GIS data preparation and eight-model automation through network analysis and public Web GIS visualization. It demonstrates how GIS can support emergency-response decision making by communicating optimized routes, response time, route distance, incidents and facility locations.

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September 2026